

Three-Model Ensemble — Model Card

TD Bank 5-day excess return vs. XFN financials ETF · XGBoost 3-class · Walk-forward validated Apr 2023 – Apr 2026

56.6%

Directional Accuracy

98.2%

Directional Call Rate

13

Walk-forward Folds

21

Total Features

Performance

Aggregate — All 13 Folds

Model	Call Rate	Dir. Accuracy	AUC (+1)	AUC (−1)
Price	0.972	0.530	0.508	0.530
Transcript	0.993	0.550	0.547	0.608
News	0.996	0.536	0.564	0.557
Hard Majority Vote	0.982	0.566	—	—

Per-Fold Detail (Hard Majority Vote)

Random baseline = 0.50. Green ≥ 0.65 (strong), yellow 0.45–0.52 (weak), red < 0.45 (below chance).

Fold	Period	Call Rate	Dir. Accuracy	Signal
1	Mar–Jun 2023	0.968	0.361	Below chance
2	Jun–Sep 2023	0.951	0.694	Strong
3	Sep–Dec 2023	1.000	0.538	Moderate
4	Dec 2023–Mar 2024	0.968	0.657	Strong
5	Mar–Jun 2024	1.000	0.572	Moderate
6	Jun–Sep 2024	0.936	0.594	Moderate
7	Sep–Dec 2024	1.000	0.686	Strong
8	Dec 2024–Mar 2025	0.985	0.388	Below chance
9	Mar–Jun 2025	1.000	0.459	Weak
10	Jun–Sep 2025	1.000	0.663	Strong
11	Sep–Dec 2025	0.968	0.457	Weak
12	Dec 2025–Mar 2026	0.985	0.691	Strong
13 *	Mar–Apr 2026	1.000	0.600	Moderate
Mean	—	0.982	0.566	—

* Fold 13: partial quarter, 20 trading days only. Strong folds align with TD-specific, macro-calm regimes; weak folds align with sector-wide shocks (SVB collapse Mar 2023, tariff escalation Dec 2024–Jun 2025).

Features

Three non-overlapping information sources. Each sub-model draws exclusively from one signal type.

Model 1 — Price6 features

Feature	Measures
td_return_5d	5-day return
td_return_20d	20-day return
td_vs_xfn_5d	5d return vs sector
td_vs_xfn_20d	20d return vs sector
td_dist_52w_high_v2	Distance from 52-week high
td_volatility_20d	20d daily std dev

Momentum, sector-relative performance, drawdown positioning, volatility regime.

Model 2 — Transcript10 features

Feature	Measures
exec_tone_ffill	CEO + CFO sentiment
transcript_analyst_qa_sentiment_mean_ffill	Analyst Q&A tone
framing_gap_ffill	Verbal vs filing divergence
topic_guidance_share_ffill	Guidance topic share
topic_guidance_sentiment_ffill	Guidance tone
topic_regulatory_AML_share_ffill	AML topic share
topic_regulatory_AML_sentiment_ffill	AML tone
topic_credit_quality_share_ffill	Credit quality share
topic_credit_quality_sentiment_ffill	Credit quality tone
days_since_call	Days since last earnings call

Executive tone, analyst reception, and the three topics most material to TD's risk profile.

Model 3 — News5 features

Feature	Measures
news_sent_mean_7d	7-day sentiment mean
news_sent_mean_30d	30-day sentiment mean
news_count_7d	Article count (7d)
news_count_30d	Article count (30d)
days_since_last_news	Days since last article

Short + medium window sentiment and volume intensity. Volume burst signals risk independently of tone.

Training & Hyperparameters

Setup

Parameter	Value
Objective	3-class multiclass (multi:softprob)
Training window	504 trading days, rolling
Leakage gap	5 trading days
Retraining cadence	Quarterly
Ensemble method	Hard majority vote (ties → 0)

Grid Search Space

Parameter	Range
max_depth	2, 3, 4
min_child_weight	10, 20, 30
n_estimators	50, 80, 120
reg_alpha (L1)	0.0, 0.2
reg_lambda (L2)	1.0, 2.0

108 configs × 3 models. Folds 4–10 used for tuning. Criterion: 0.6 × directional accuracy + 0.4 × macro F1.

Best Parameters (tuned 2026-04-23)

Model	max_depth	min_child_weight	n_estimators	reg_alpha	reg_lambda
Price	3	30	120	0.0	1.0
Transcript	4	20	80	0.2	1.0
News	2	30	50	0.2	2.0

Ensemble Logic

Price	Transcript	News	Hard Vote Result
+1	+1	-1	+1 (majority)
-1	0	-1	-1 (majority)
+1	-1	0	0 (three-way split → neutral)

Hard majority vote outperforms soft-prob averaging — models share strong directional signal but are not well-calibrated in probability space.

Limitations

- Data

Single stock, limited rows — ~800 rows per fold constrains model complexity and limits exposure to rare high-impact events.
- Scope

Directional only — predicts direction vs. XFN, not magnitude. Position sizing requires a separate model.
- Regime

Macro-blind — in market-wide shock regimes all three feature sets are TD-specific and cannot capture cross-asset forces (tariff escalation, banking system stress).
- News

Coverage gaps — limited to TD press releases; transcript features are forward-filled between calls and become significantly stale before the next earnings event.